

RESEARCH ARTICLE

Exotic species explain plant functional trait differences between seed mixes, restored and reference prairies

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Abstract

Questions: Are community-weighted plant functional traits related to the 'fast-slow' growth rate continuum different in seed mixes, restored and reference prairies? If so, then what are the traits that explain differences? Remnant prairies, which serve as reference sites for restorations, often have higher plant species diversity and a lower abundance of exotic (non-native) species than their restoration counterparts. They may differ in their community-weighted functional traits as well, although this is poorly studied.

Location: Tallgrass prairies and abandoned fields in Iowa, United States, North America.

Methods: We compared functional traits between remnant prairies, restoration seed mixes, and established restorations to determine if community-weighted trait values ($\sum p_i \times \text{traits}$) are comparable. Relative abundance was determined by previous sampling of 93 restorations, 48 seed mixes and five remnant sites. Functional traits, including leaf specific leaf area (SLA) and leaf dry matter content (LDMC), and plant height, were measured for 145 native and 39 non-native plant species, 663 plants, and 1326 leaves.

Results: Seed mixes had greater community-weighted LDMC and lower SLA than remnants, perhaps because of human selection for high grass abundance in mixes. Established restorations had lower community-weighted LDMC and higher SLA than remnants and seed mixes, and restorations had a large exotic species component. The proportional abundance of exotics in restorations explained trait differences between remnants and restorations. After taking into account functional groupings, perennial exotic species, on average, had higher SLA and lower LDMC than perennial native species. Remnants and established restorations had similar plant heights, and height was not significantly different between native and exotic species.

Conclusions: Our results indicate that seed mixes differ from remnants in a manner that was the opposite of the difference between seed mixes and established restorations. Although seed mixes favored species with slower growth rate trait compositions, the species that established in restored prairies tended to have traits associated with faster growth rates. We recommend developing prairie restoration seed mixes with a greater proportion of forbs, especially those with lower LDMC and higher SLA.

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